

Error Back Propagation

Introduction

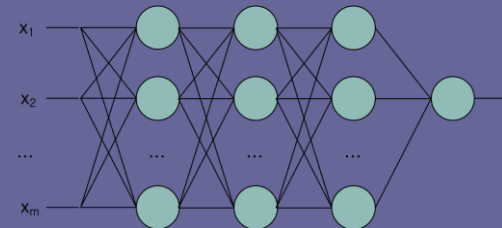
- Preparation for Learning
 - Given input-output data of the target function to learn
 - Given structure of network (# of nodes in hidden layer)
 - Randomly initialized weights

Given
data

$$\begin{aligned}
 &(x_{11}, x_{12}, x_{13}, \dots, t_1) \\
 &(x_{21}, x_{22}, x_{23}, \dots, t_2) \\
 &\dots \\
 &(x_{n1}, x_{n2}, x_{n3}, \dots, t_n)
 \end{aligned}$$

Neural
network

An Initial Network



Introduction

- Learning Algorithm
 - Update connection weights

Given data

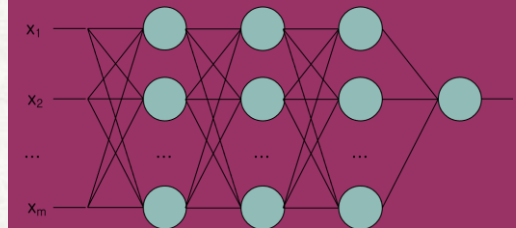
$(x_{11}, x_{12}, x_{13}, \dots, t_1)$
 $(x_{21}, x_{22}, x_{23}, \dots, t_2)$
 $\dots$
 $(x_{n1}, x_{n2}, x_{n3}, \dots, t_n)$



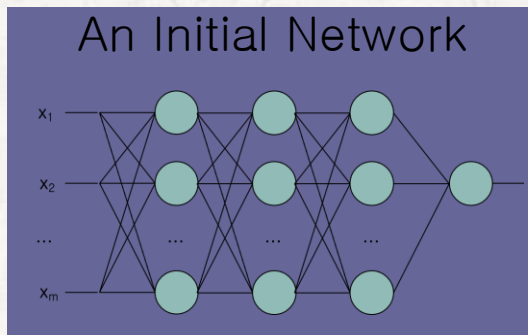
Learning Algorithm



A Network after learning the given data



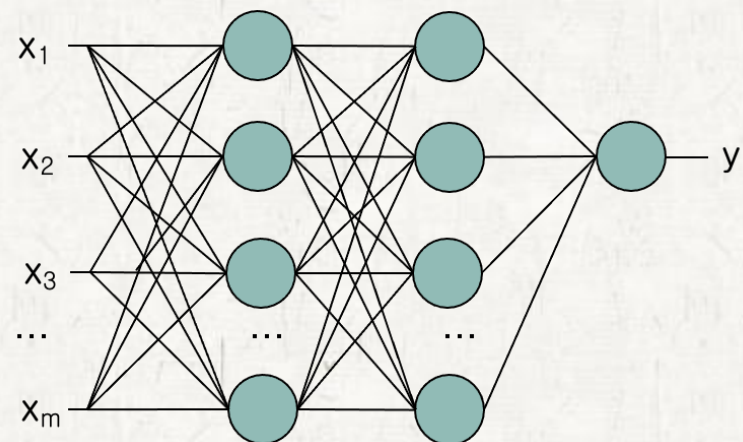
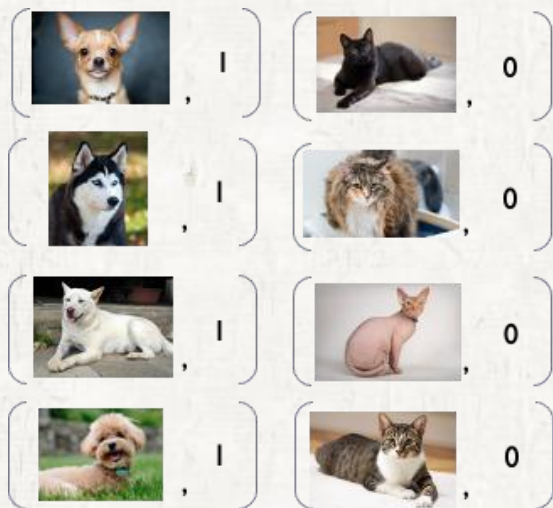
Neural network



What are different??

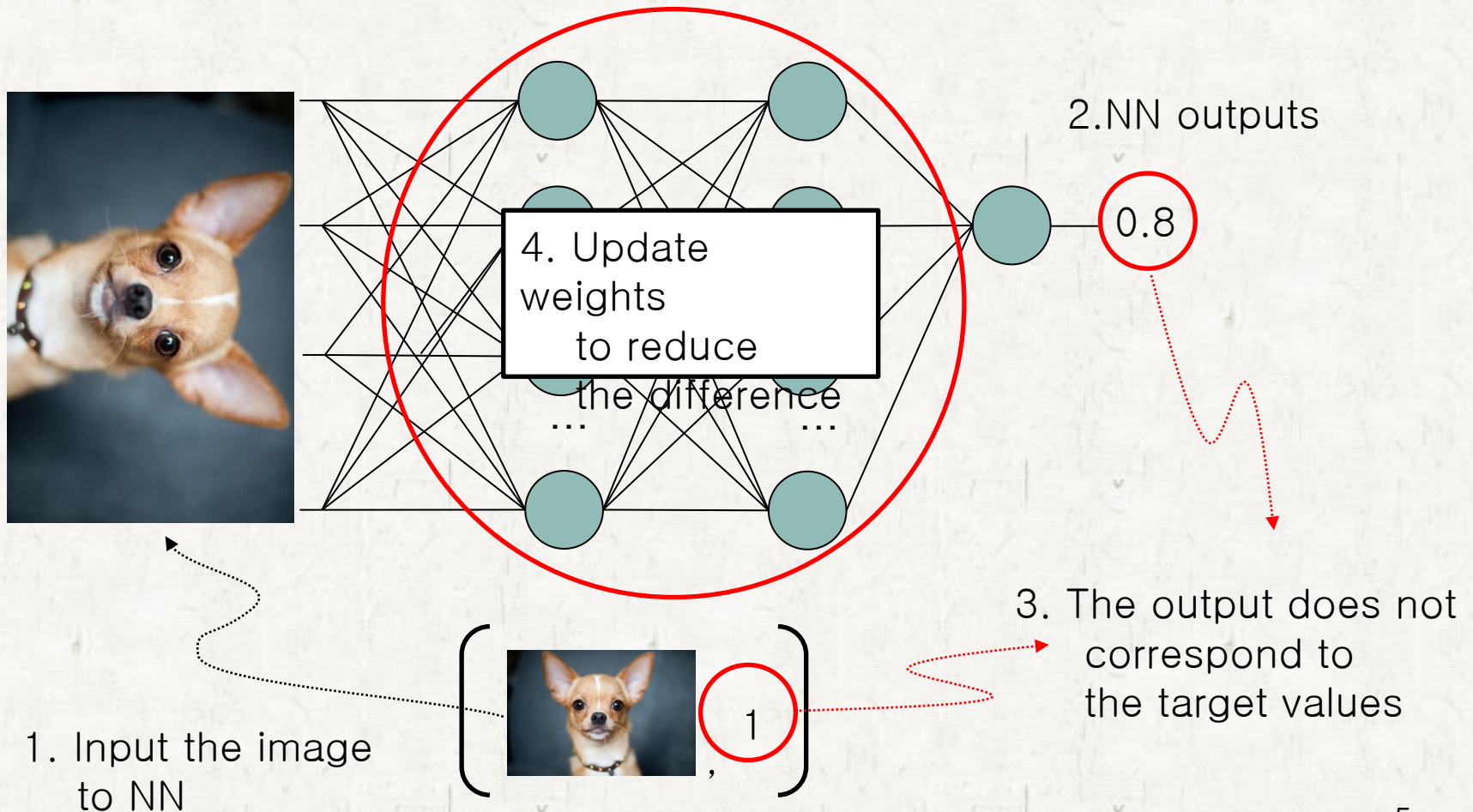
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Learning Algorithm



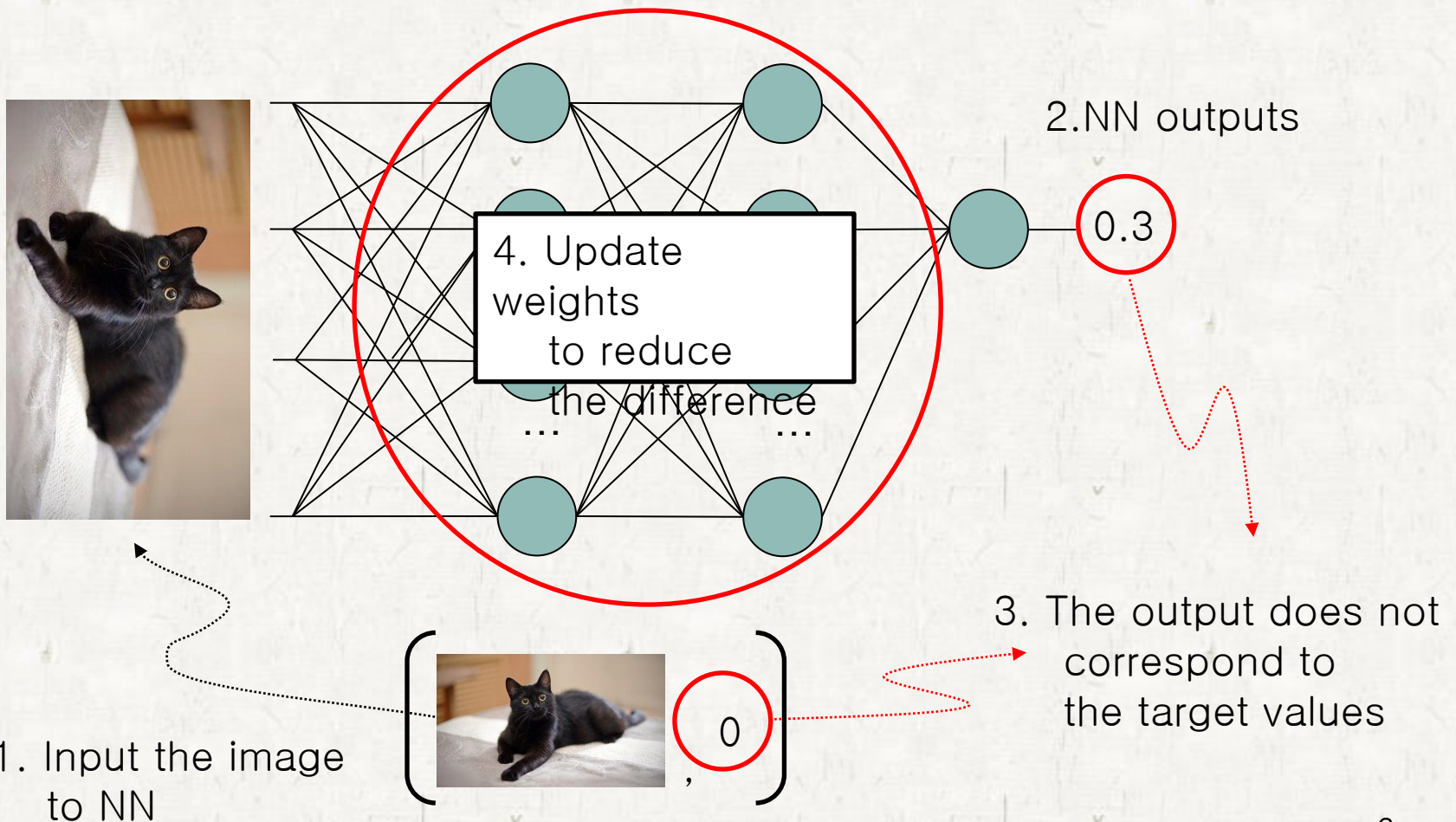
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Learning Algorithm



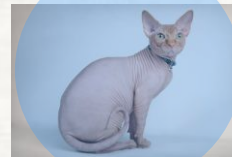
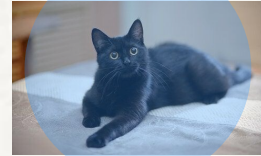
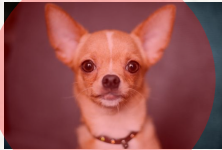
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Learning Algorithm



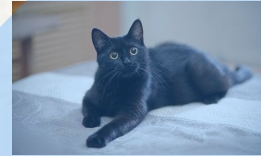
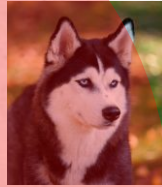
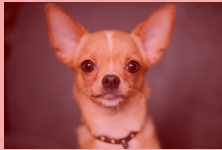
Introduction

- Learning Algorithm: 배운 것을 외우게 됨



Introduction

- Learning Algorithm: 배운 것을 일반화하게 됨

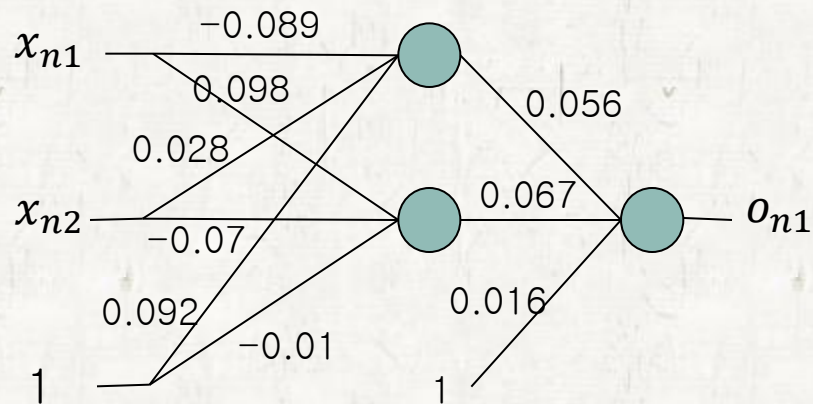


Example of Error Back Propagation

Example : XOR

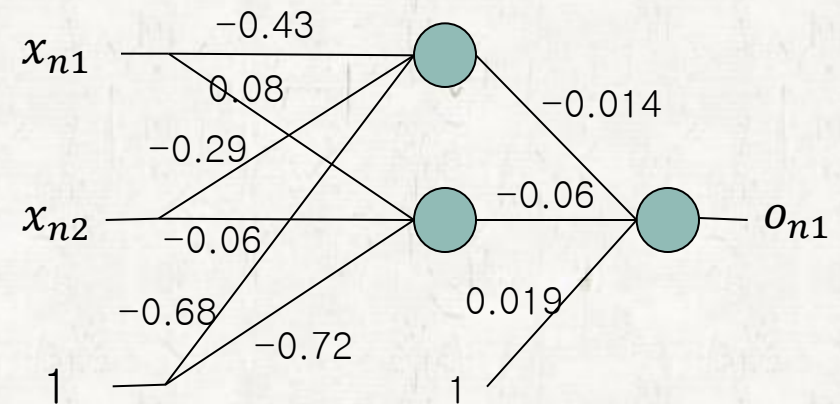
Iteration : 0

x_{n1}	x_{n2}	t_{n1}	o_{n1}
1	1	0	0.52
1	0	1	0.50
0	1	1	0.52
0	0	0	0.55



Iteration : 1000

x_{n1}	x_{n2}	t_{n1}	o_{n1}
1	1	0	0.50
1	0	1	0.48
0	1	1	0.50
0	0	0	0.52

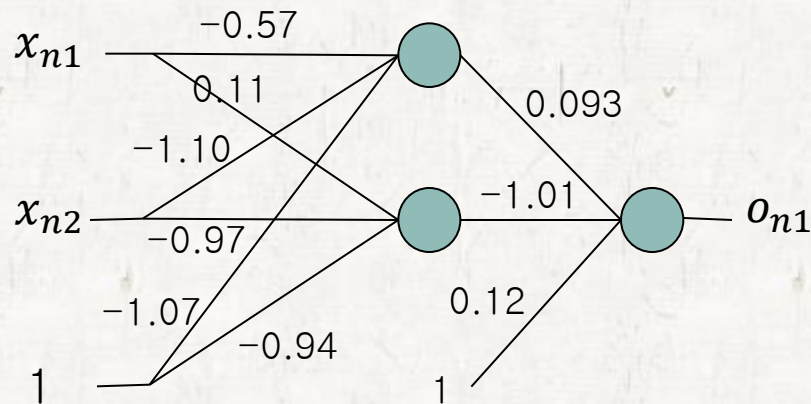


Example of Error Back Propagation

Example : XOR

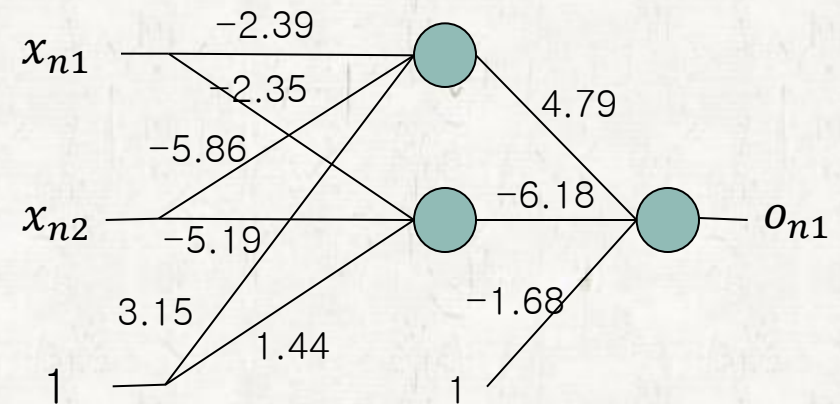
Iteration : 2000

x_{n1}	x_{n2}	t_{n1}	o_{n1}
1	1	0	0.53
1	0	1	0.48
0	1	1	0.50
0	0	0	0.48



Iteration : 3000

x_{n1}	x_{n2}	t_{n1}	o_{n1}
1	1	0	0.30
1	0	1	0.81
0	1	1	0.81
0	0	0	0.11

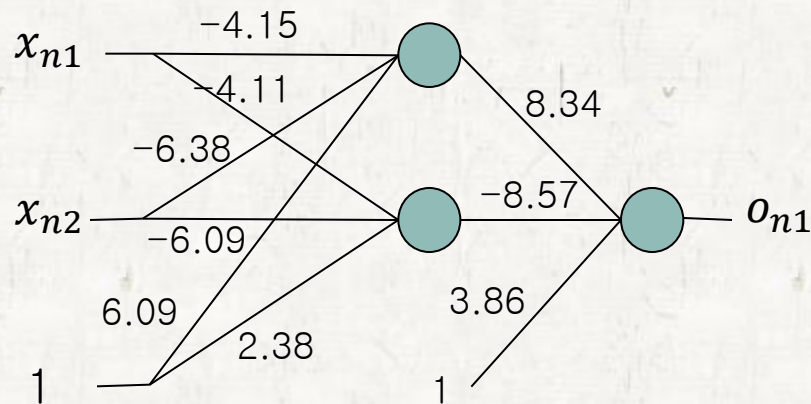


Example of Error Back Propagation

Example : XOR

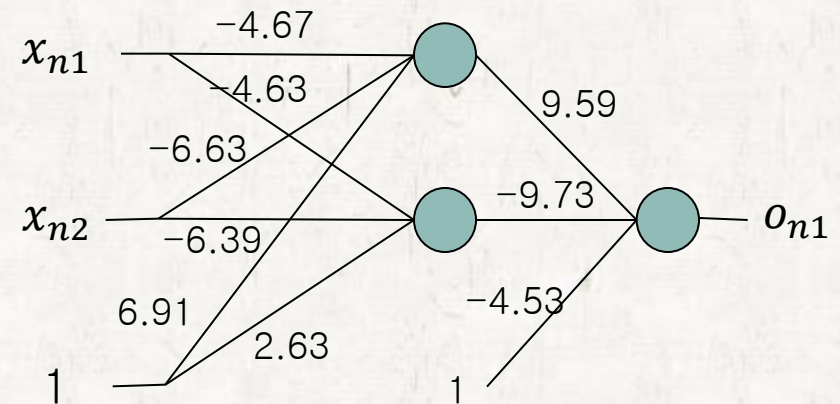
Iteration : 5000

x_{n1}	x_{n2}	t_{n1}	o_{n1}
1	1	0	0.05
1	0	1	0.96
0	1	1	0.96
0	0	0	0.03



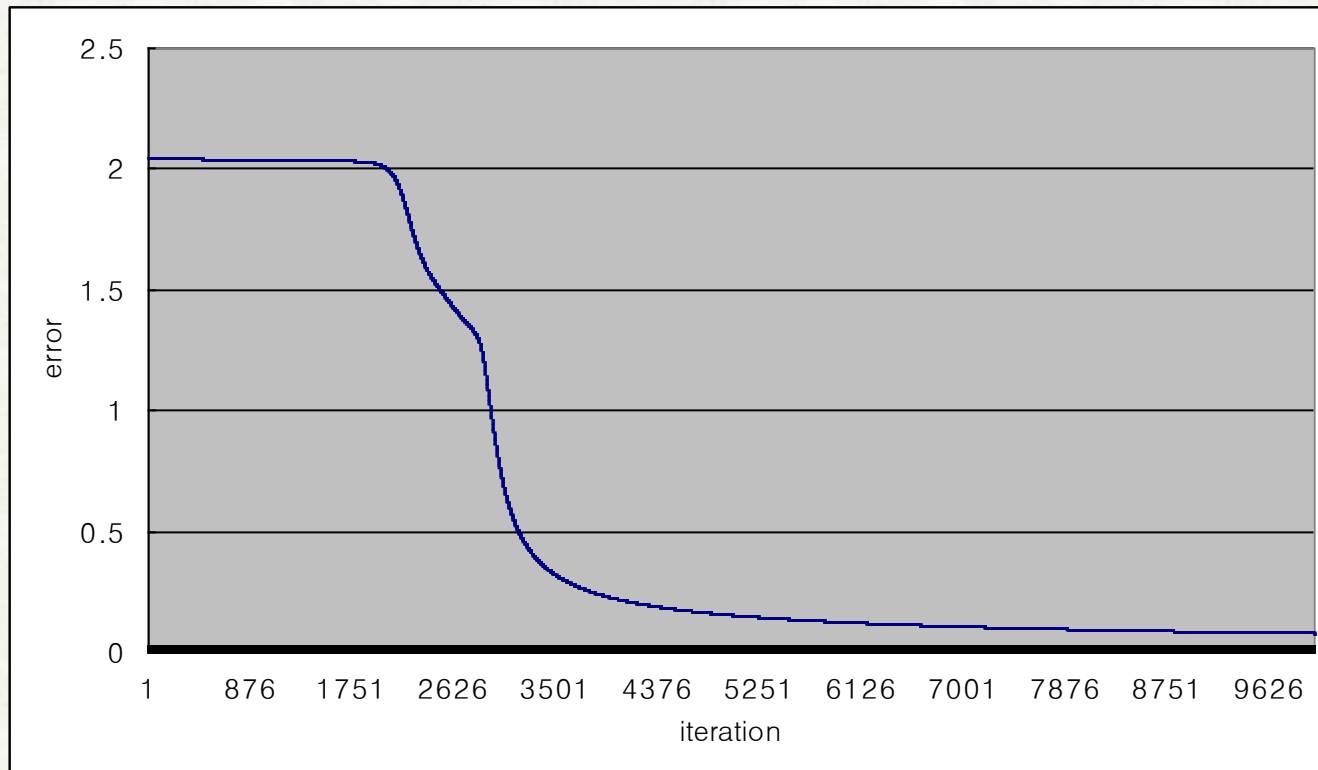
Iteration : 10000

x_{n1}	x_{n2}	t_{n1}	o_{n1}
1	1	0	0.02
1	0	1	0.98
0	1	1	0.98
0	0	0	0.02



Example of Error Back Propagation

- Example : XOR
- Error graph

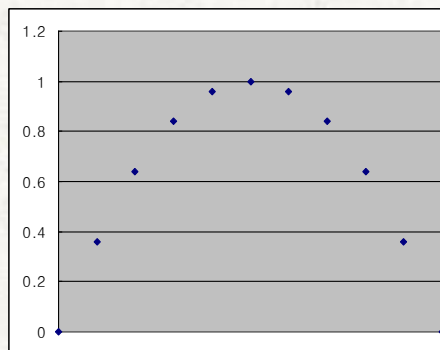
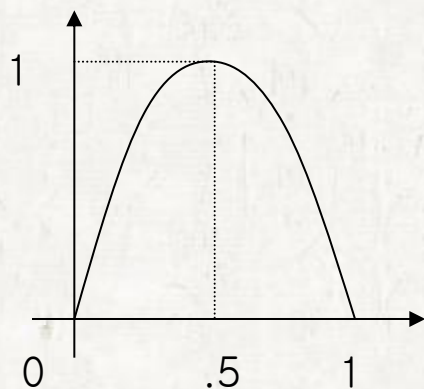


Example of Error Back Propagation

● Example2 :

- Hidden nodes : 4
- Iteration : 500,000
- Learning rate : 0.7

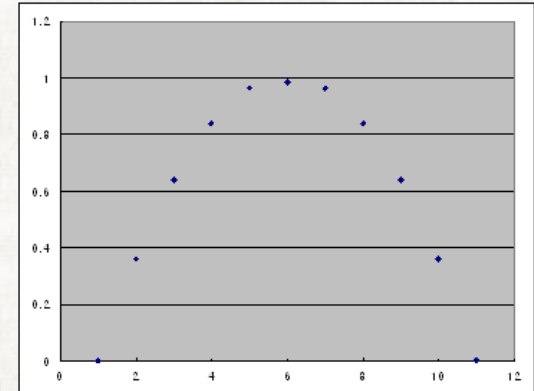
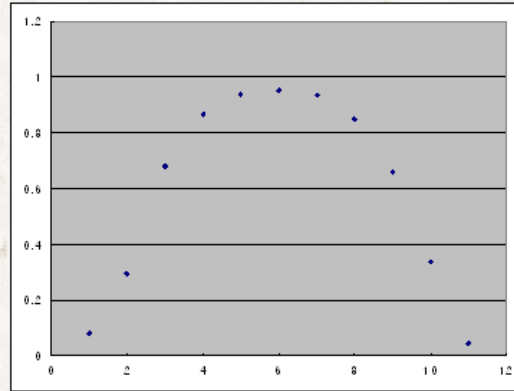
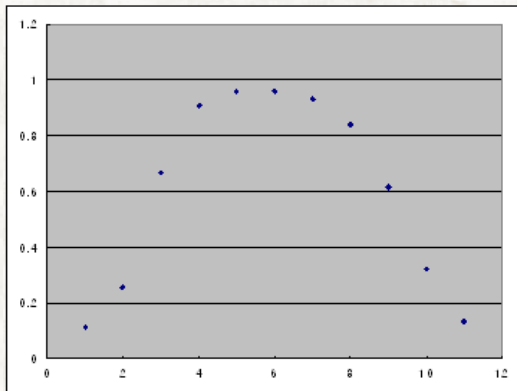
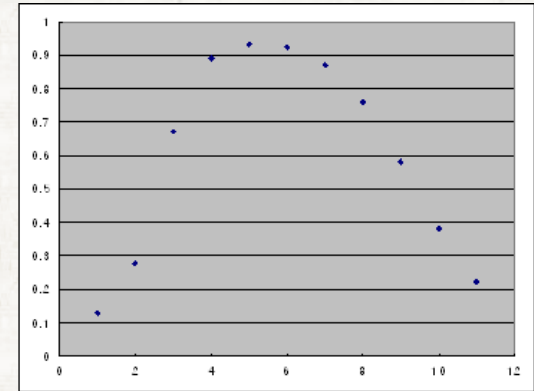
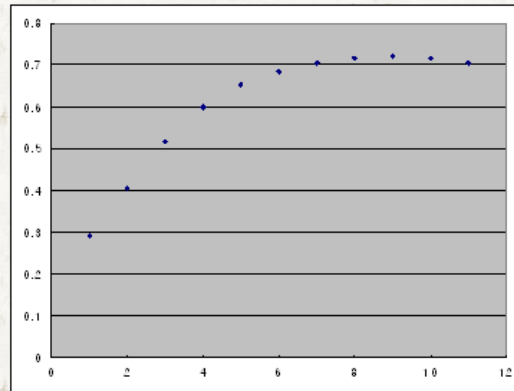
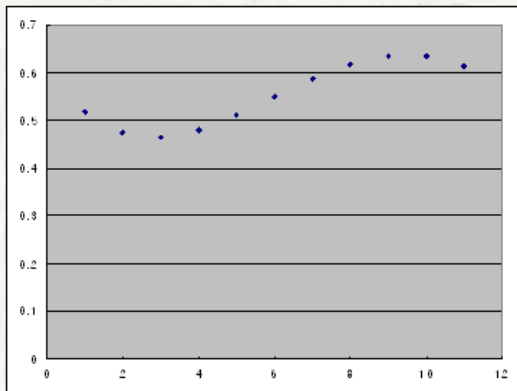
$$f(x) = 4x*(1-x)$$



Input	Output
0.00	0.00
0.10	0.36
0.20	0.64
0.30	0.84
0.40	0.96
0.50	1.00
0.60	0.96
0.70	0.84
0.80	0.64
0.90	0.36
1.00	0.00

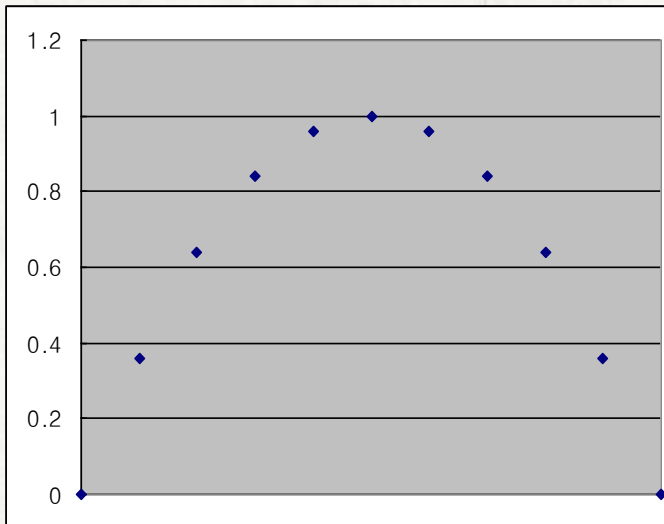
Example of Error Back Propagation

Example2

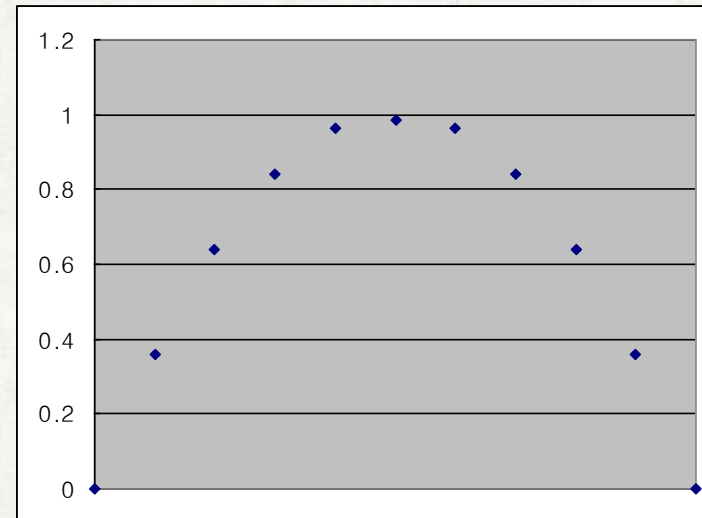


Generalization and Overfitting (1)

- We gave only 11 points
 - A NN learned only that 11 points



Training data

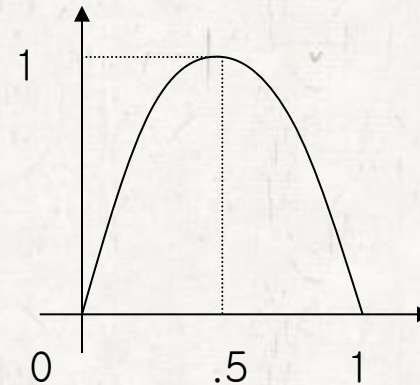
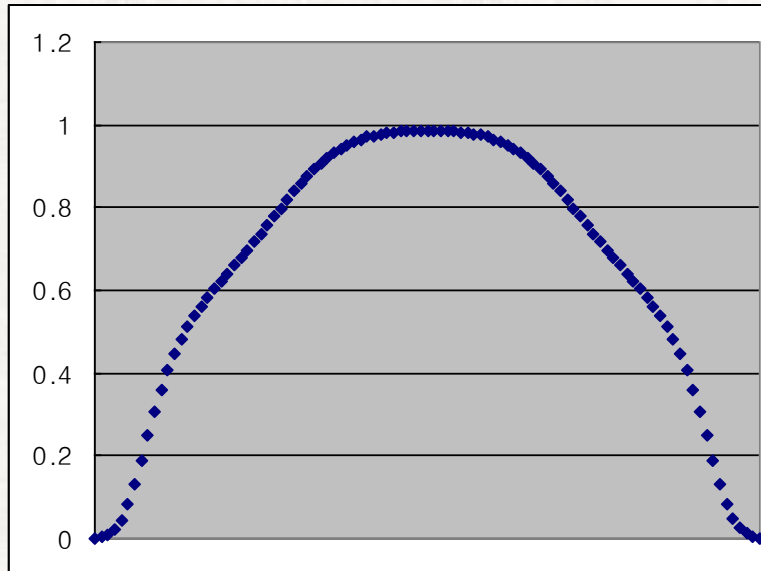


Training result

- Can the NN answer to the un-learned points?

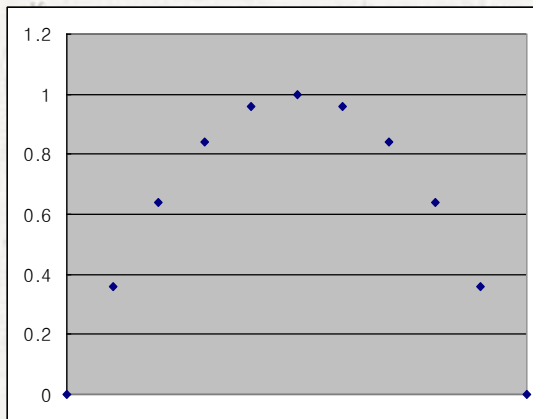
Generalization and Overfitting (2)

- Yes, NNs generalize what they have learned

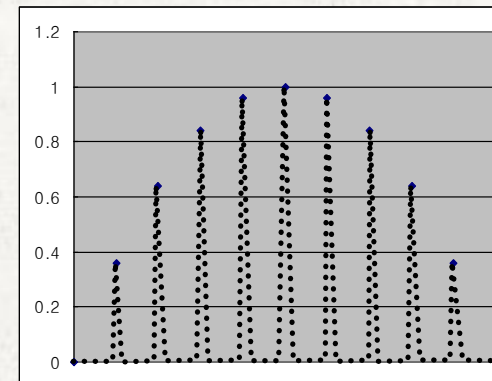
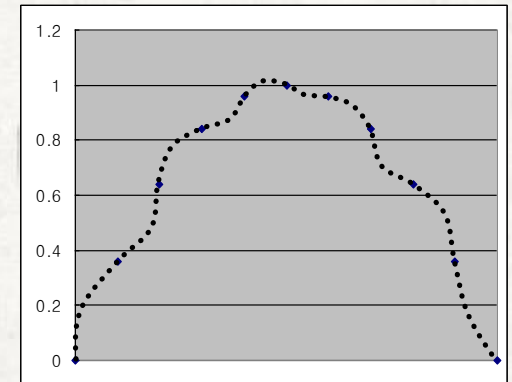
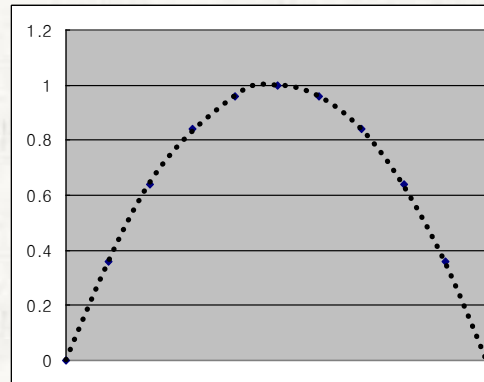


Generalization and Overfitting (3)

Which one is better?

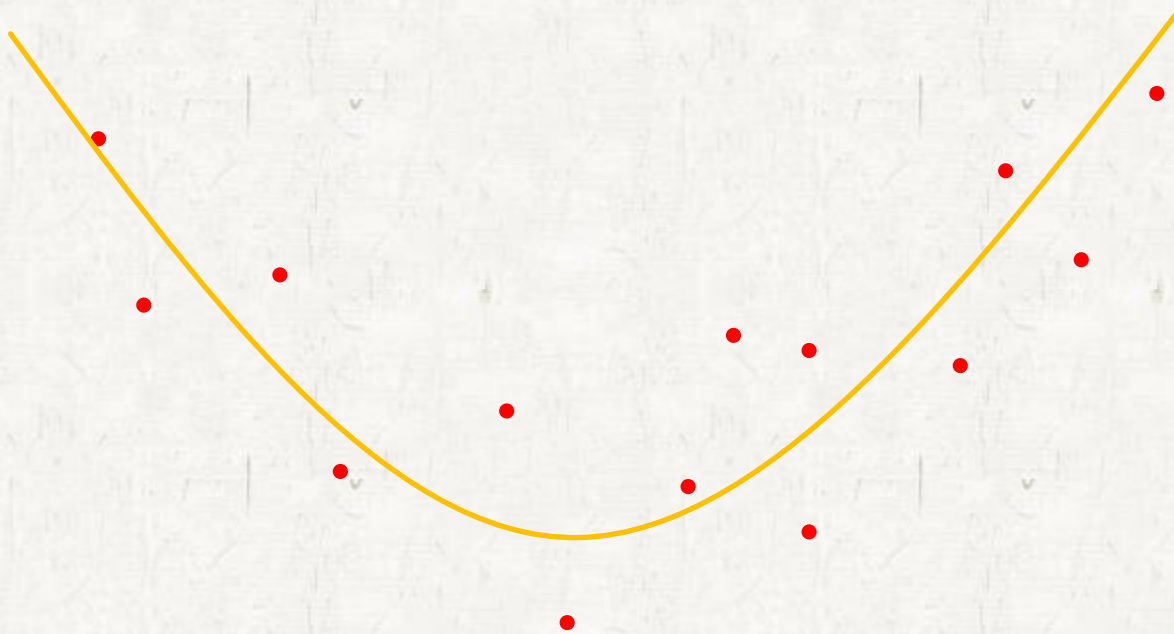


Training data



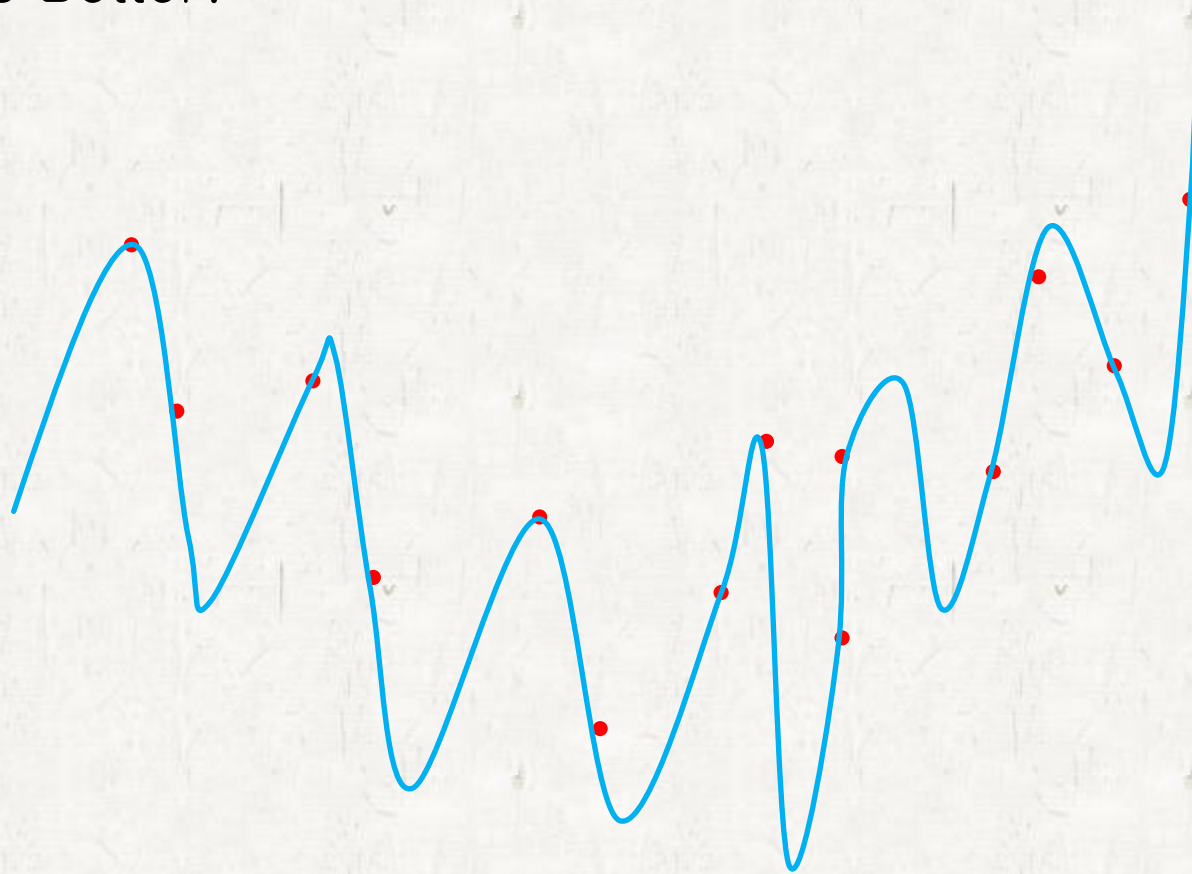
Generalization and Overfitting (4)

- Which is Better?



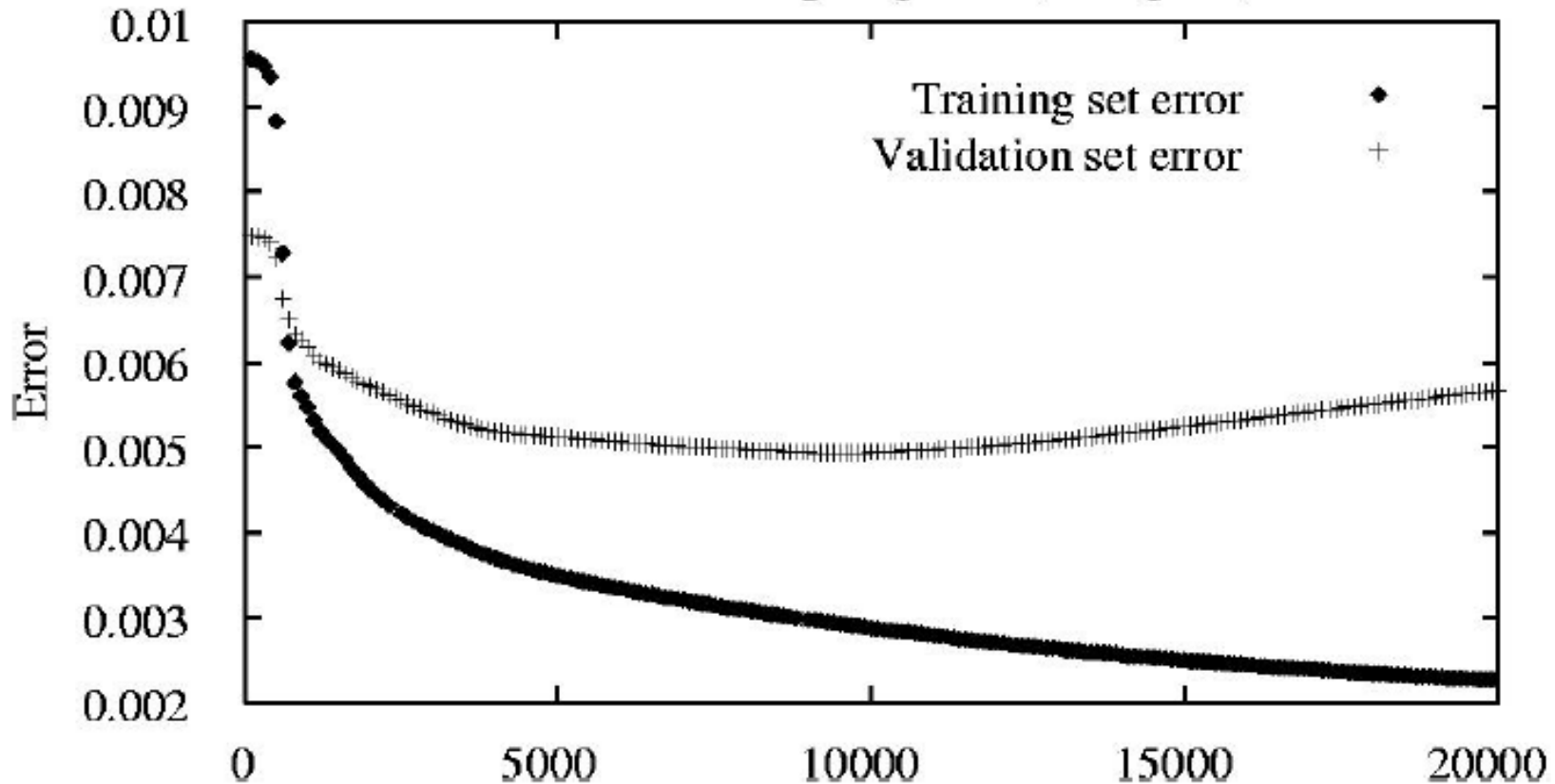
Generalization and Overfitting (5)

- Which is Better?



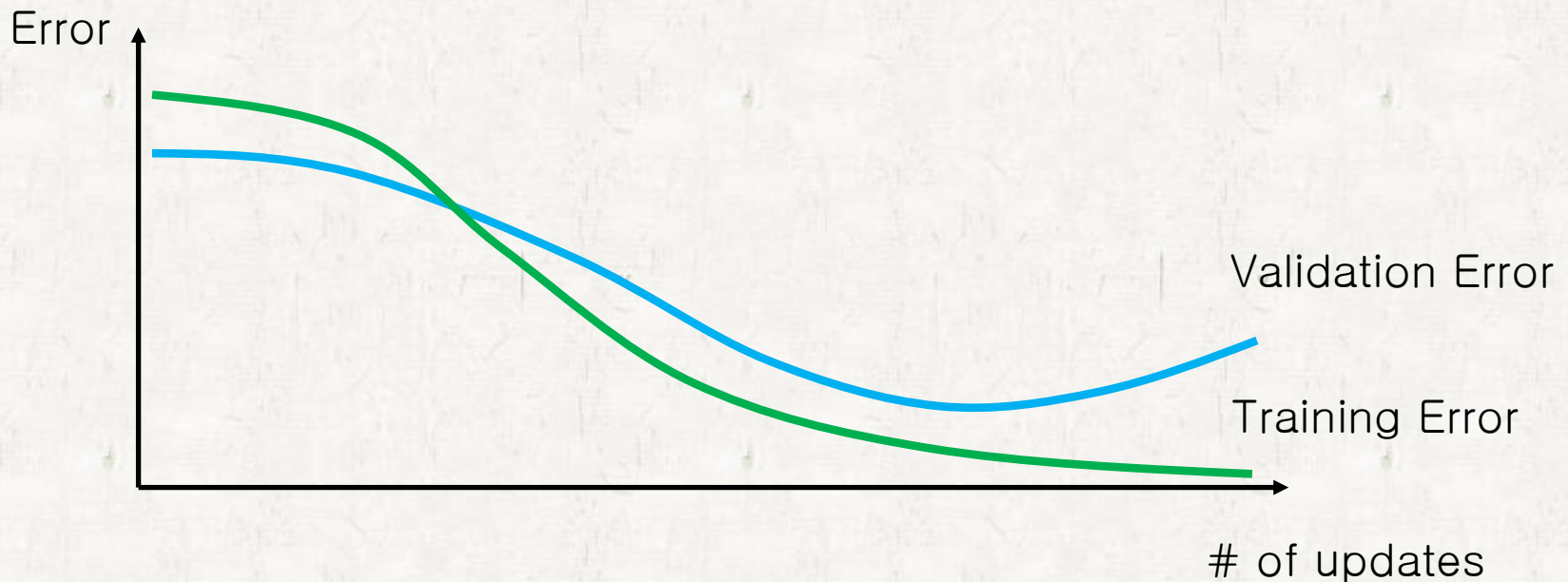
Generalization and Overfitting (6)

Error versus weight updates (example 1)



Generalization and Overfitting (7)

Early Stopping



Generalization and Overfitting (8)

- To increase generalization accuracy
 - Find the optimal number of neurons
 - Find the optimal number of training iterations
 - Use regularization
 - Use more training data